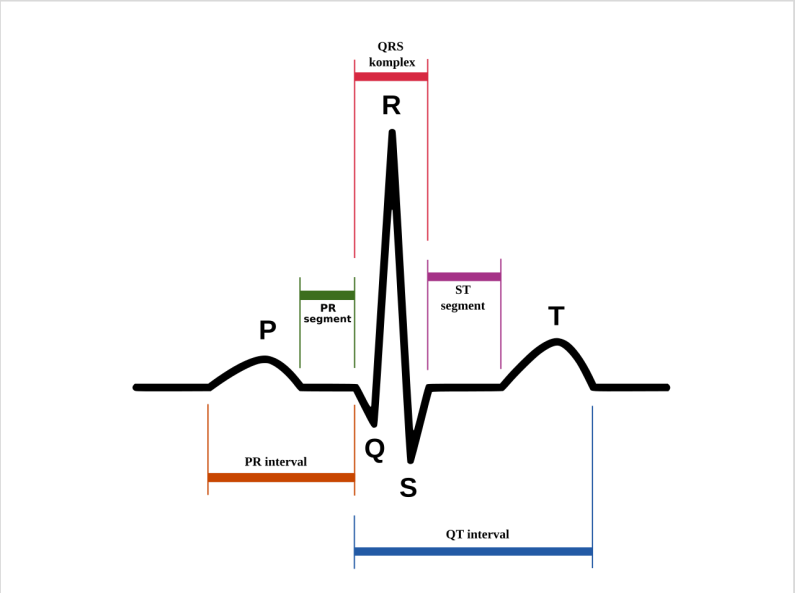
ECG Signal Simulator in MATLAB

This project involves the development of a MATLAB-based simulator that generates realistic electrocardiogram (ECG) signals. The program models the key components of a heartbeat, including P, Q, R, S, T, and U waves, and allows users to adjust parameters such as heart rate, wave amplitudes, durations, and intervals. The simulation produces a continuous, dynamic ECG waveform, providing a visual representation of cardiac activity.



Labeled ECG Waveform

P wave

- The first small upward deflection on an ECG.
- Represents **atrial depolarization**, which is the electrical activity that triggers the atria (upper heart chambers) to contract.
- It occurs just before the ventricles activate.

PR interval / segment

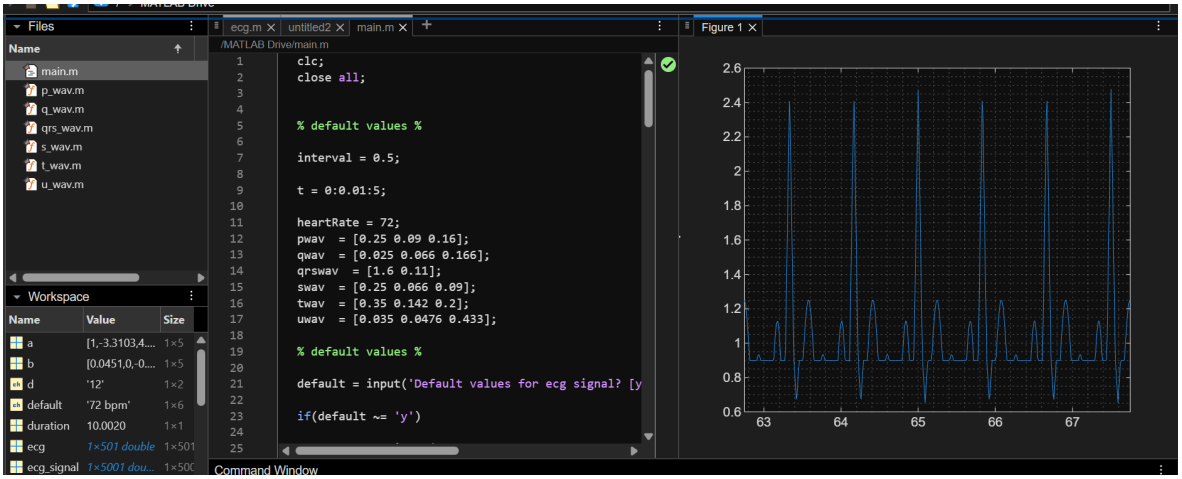
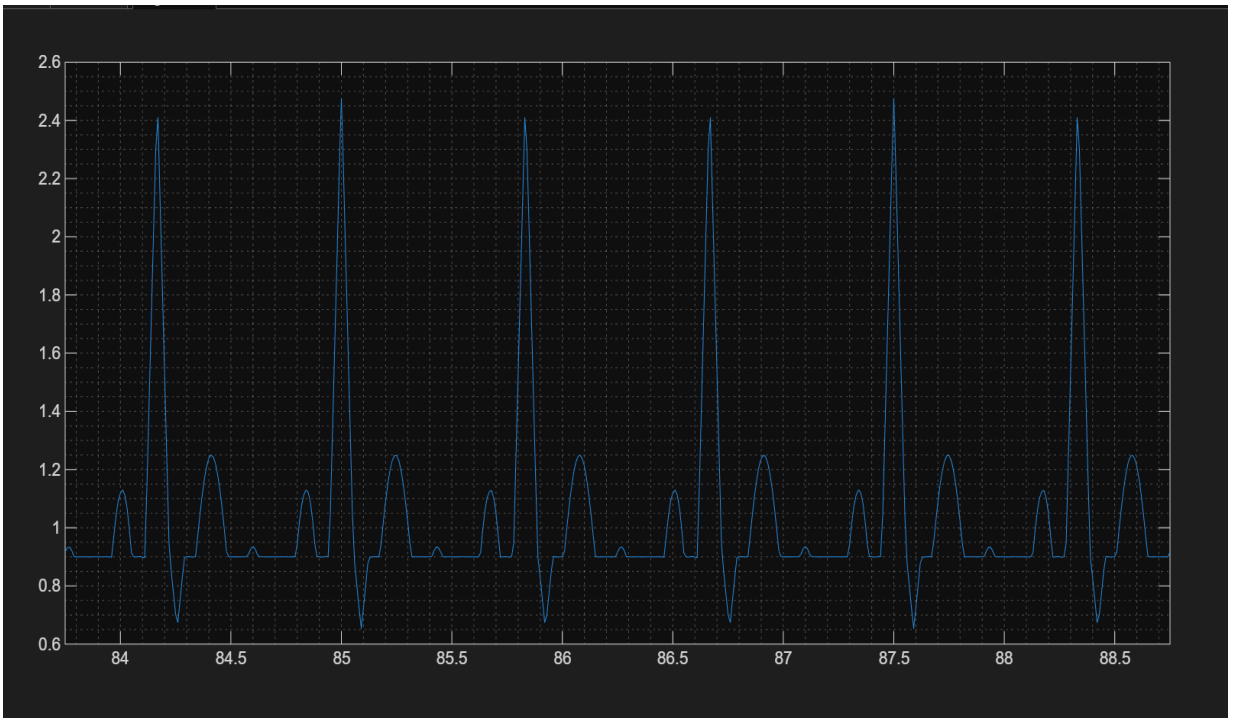
- The time between the start of the P wave and the start of the QRS complex.
- It reflects the delay as the electrical impulse passes through the AV node before ventricular activation.

QRS complex

- A group of three waves (Q, R, S) that together represent **ventricular depolarization** , the electrical activation of the ventricles (lower heart chambers).
    - **Q wave**: first small downward deflection after the P wave.
    - **R wave**: the first large upward spike.
    - **S wave**: downward deflection after the R wave.
- Because the ventricles are large muscle chambers, this part of the signal is the most visible and sharpest on ECG.

T wave

- Follows the QRS complex and appears as a broader upward curve.
- Represents **ventricular repolarization**, which is the process of the ventricles resetting electrically in preparation for the next beat.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